Bank Customer Analysis Dashboard

1. Problem Statement

Customer churn is a major challenge for banks, as retaining existing customers is more cost-effective than acquiring new ones. Understanding why customers leave and which customer segments are at higher risk of churn is essential for improving customer retention strategies.

The objective of this project is to analyze bank customer data to:

- Identify churn patterns
- Detect high-risk customer segments
- Provide actionable insights to help reduce customer attrition

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2. Dataset Overview

The dataset used for this project contains detailed information about bank customers and their churn behavior. It is designed to help analyze customer demographics, financial attributes, and activity patterns to understand why customers leave the bank.

Dataset Size

Total Records: ~10,000 customers

Format: CSV (Comma-Separated Values)

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3. Tools Used

The following tools were used in this project:

- **Power BI** – Data modeling, visualization, and interactive dashboard creation
- **Excel** – Initial data review and basic preprocessing
- **CSV** – Data source

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4. Key Metrics (KPIs)

The dashboard focuses on the following key performance indicators:

- **Total Customers**
- **Churn Rate (%)**
- **Active vs Inactive Customers**
- **Customer Distribution by Gender**
- **Customer Distribution by Country**

These KPIs provide a high-level view of customer behavior and churn trends.

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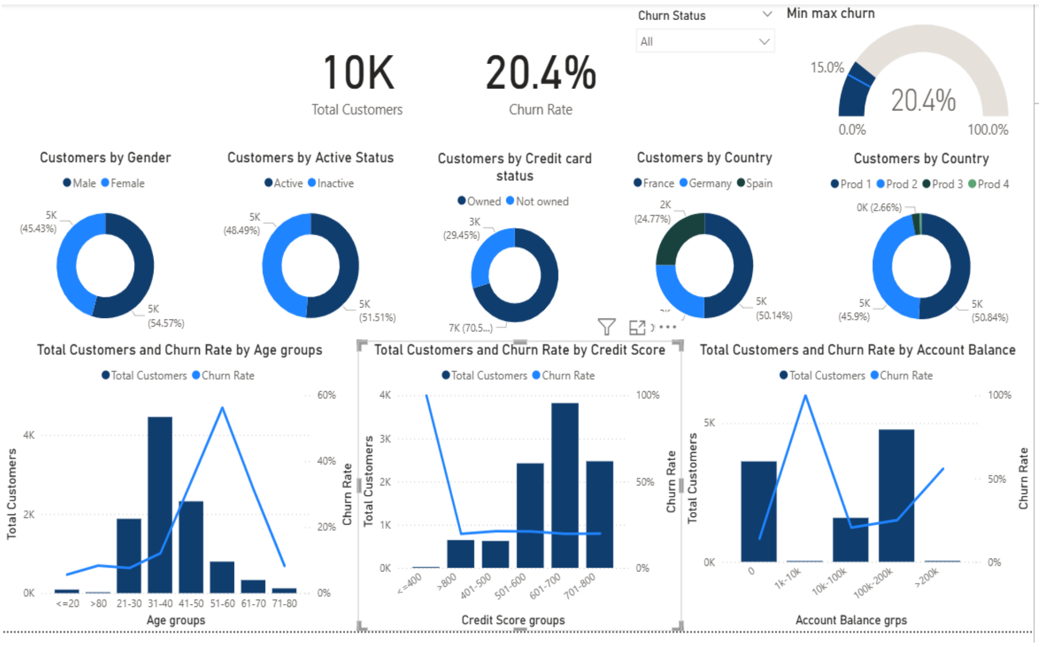
5. Power BI Dashboards

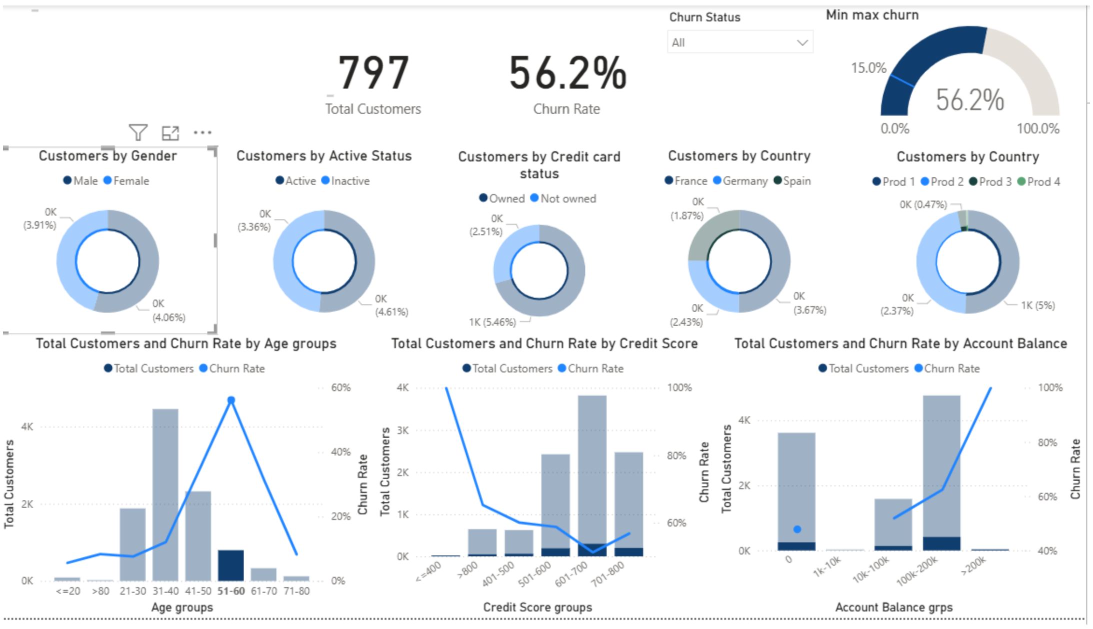
Bank customer churn overview dashboard

Shows an overall view of customer churn and key metrics.



churn_high_risk_segment dashboard

Shows customers who are more likely to leave the bank.



6. High-Risk Customer Segment Analysis

Age Group Analysis

The analysis shows that customers aged **51–60** form the **highest churn risk segment**.

- **Churn Rate:** Approximately 56%
- **Observation:** Churn increases significantly in this age group compared to younger customers

A filtered dashboard view highlights this high-risk segment, helping stakeholders focus on customers most likely to churn.

Credit Score Analysis

- Customers with **low credit scores** have higher churn rates
- Churn risk decreases as credit score improves

This indicates a strong relationship between customer financial stability and retention.

Account Balance Analysis

- Customers with **low or zero account balances** show higher churn
- Customers with higher balances are generally more stable but still require engagement

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7. Key Insights

- Customers aged **51–60** have the highest churn rate
- Low credit score customers are more likely to churn
- Inactive customers churn more frequently than active customers
- Low account balance is a strong indicator of churn risk

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8. Business Recommendations

Based on the insights from the analysis, the following actions are recommended:

1. **Targeted Retention Strategies**
    - Design personalized offers for customers aged 51–60
    - Introduce loyalty benefits and relationship-based banking programs
2. **Support for Low Credit Score Customers**
    - Provide financial guidance and flexible banking products
    - Improve customer engagement to build long-term trust
3. **Reactivation of Inactive Customers**
    - Launch re-engagement campaigns for inactive users
    - Encourage digital banking usage and personalized communication
4. **Balance-Based Incentives**
    - Offer incentives for maintaining minimum account balances
    - Promote value-added services for low-balance customers